

# Guide to the Chi-Square Test ( $\chi^2$ )

## Test of Independence & Categorical Analysis

The Chi-Square test of independence determines if there is a significant relationship between two categorical variables. It compares what we **Observed** in real life versus what we would **Expect** if there were no relationship at all.

### 1. The Scenario: Coffee Shop Habits

Research Question: Does "Customer Type" influence "Drink Choice"?

| Category     | Coffee | Tea | Row Total         |
|--------------|--------|-----|-------------------|
| Student      | 20     | 30  | 50                |
| Professional | 40     | 10  | 50                |
| Col Total    | 60     | 40  | 100 (Grand Total) |

### 2. The Formula

$$\chi^2 = \sum [ (O - E)^2 / E ]$$

O = Observed frequency | E = Expected frequency |  $\Sigma$  = Sum of all cells

### 3. Step-by-Step Calculation

#### Step A: Calculate Expected Values (E)

Formula:  $E = (\text{Row Total} \times \text{Column Total}) / \text{Grand Total}$

- Student + Coffee:  $(50 * 60) / 100 = 30$

- Student + Tea:  $(50 * 40) / 100 = 20$
- Prof + Coffee:  $(50 * 60) / 100 = 30$
- Prof + Tea:  $(50 * 40) / 100 = 20$

### Step B: Apply Formula for Each Cell

1. Student/Coffee:  $(20 - 30)^2 / 30 = 3.33$
2. Student/Tea:  $(30 - 20)^2 / 20 = 5.00$
3. Prof/Coffee:  $(40 - 30)^2 / 30 = 3.33$
4. Prof/Tea:  $(10 - 20)^2 / 20 = 5.00$

### Step C: Sum the Values

$$\chi^2 = 3.33 + 5.00 + 3.33 + 5.00 = 16.66$$

## 4. Final Conclusion

**Degrees of Freedom (df):**  $(2-1) * (2-1) = 1$

**Critical Value:** (at  $\alpha=0.05$ ,  $df=1$ ) = 3.84

If  $\chi^2 > \text{Critical}$ : Reject Null (Relationship is REAL)

**Result:** Since  $16.66 > 3.84$ , we **REJECT** the null hypothesis. There is a statistically significant relationship: Professionals prefer coffee at a significantly higher rate than students.